INFORMATION TO KEEP WITH YOU

My complete treatment handbook

Pembrolizumab + pemetrexed + carboplatin

Understand your treatment. Know what to expect. Keep your plan to hand.

Displayed calendar
21 days
See the active treatment phase

Treatment stages
2
Explained in the treatment reference

Regimen source
NCCP 00568
Version 5

> Day 1 every 21 days. Up to four cycles with all three medicines, then pembrolizumab and pemetrexed may continue without carboplatin.

Treatment stages

01 **Induction · cycles 1–4**
Three medicines

02 **Maintenance · after induction**
Two medicines

Your team confirms the active stage and dates. The full treatment guide explains each step.

Inside your handbook

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My name _____

Hospital / oncology team _____

Use the urgent instructions and contact numbers in your treatment guide. Your team must replace example numbers before patient use. Your individual treatment plan takes precedence.

CHAPTER 01

Understanding your treatment

IN THIS CHAPTER

Why this treatment is offered, what its medicines do, and how the agreed plan fits your care.

Pembrolizumab + pemetrexed + carboplatin
National Cancer Control Programme (NCCP) 00568

When to get help

Emergency: call 112 / 999

Severe breathing difficulty, collapse, seizure or severe chest pain: call emergency services. Do not wait for an online answer.

Contact your oncology service immediately

Temperature 37.5°C or higher, 36°C or lower, shivering or suddenly feeling unwell — even with a normal temperature. Contact the oncology team promptly for fever or feeling acutely unwell, new breathlessness or cough, severe diarrhoea, dehydration, chest pain, confusion, severe rash or other new symptoms. Use the emergency contact instructions given by your team.

Follow the urgent contact instructions from your treating hospital. If you cannot reach the team, seek urgent medical assessment. Outside clinic hours, use your instructed out-of-hours pathway or the emergency department.

This page does not diagnose symptoms or decide whether it is safe to stay at home. You do not need to know which medicine caused a symptom before calling.

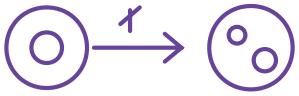
Three medicines. Different jobs.

The combination aims to control metastatic non-squamous lung cancer. Your team checks cancer markers, your health and your preferences. Chemotherapy and immunotherapy work in different ways.

IMMUNOTHERAPY

Pembrolizumab

Keytruda



Releases an immune brake

PD-1 is a control switch on T cells, a type of immune cell. When it connects with PD-L1 or PD-L2, the T cell receives a signal to reduce its activity. Some cancers use this pathway to escape immune attack. Pembrolizumab is an antibody that blocks PD-1, helping T cells recognise and attack the cancer. It does not directly poison cancer cells like chemotherapy. Releasing this brake can also allow immune cells to attack healthy organs, which explains the different side-effect pattern and why symptoms can appear after treatment stops.

Immune cell → Brake blocked → Cancer cell

What to watch for

The immune system can also inflame healthy organs. New cough, diarrhoea, unusual weakness, severe headache, thirst or a spreading rash need prompt advice. Effects can start during treatment or after it finishes.

CHEMOTHERAPY

Pemetrexed

Alimta



Disrupts deoxyribonucleic acid (the cell's genetic material) (DNA) building

Pemetrexed is an antifolate: it blocks several enzymes that cells use to turn folate-dependent materials into DNA and RNA building blocks. With fewer building blocks, cancer cells have more difficulty copying their genetic instructions and dividing. Healthy rapidly renewing tissues, including bone marrow and the mouth lining, can also be affected. That helps explain low blood counts and mouth soreness. Prescribed folic acid and vitamin B12 reduce toxicity; they are part of the treatment, not supplements to omit because the drug is an antifolate.

DNA building blocks → Supply disrupted → Cell division slows

What to watch for

Healthy tissues can be affected too: low blood counts, a sore mouth, rash and kidney problems are important. Folic acid, vitamin B12 and prescribed steroids help reduce particular risks.

CHEMOTHERAPY

Carboplatin

Platinum chemotherapy



Damages cancer-cell DNA

Carboplatin is a platinum-containing chemotherapy. Inside cells, it forms chemical attachments to DNA, including links that stop DNA strands working normally. This disrupts copying and repair and can lead to cell death. Healthy cells can also be injured, especially blood-forming cells in the bone marrow. The kidneys help remove carboplatin, so kidney function is used to calculate the dose. AUC 5 describes a target drug exposure over time; it is not a fixed dose of 5 mg.

DNA strands → Platinum links → Copying disrupted

What to watch for

Low blood counts, nausea and vomiting can occur. An allergic reaction is also possible: tell the nurse immediately if you feel flushed, itchy, tight-chested or unwell during the drip.

The drawings are simplified explanations of drug action, not anatomical pictures or a prediction of response.

Understanding your diagnosis and treatment aim

Start here if the names and abbreviations are new. You can read one section at a time and return when you need it.

What do the cancer terms mean?

Non-small-cell lung cancer (NSCLC) is a broad group of lung cancers. “Non-squamous” describes the cells seen by the pathologist and includes adenocarcinoma. “Metastatic” means the cancer has spread beyond its original site. “First-line” means the first drug treatment chosen for this stage of illness. A regimen is a planned combination and schedule of medicines.

What is the treatment trying to achieve?

The usual aims in this setting are to control cancer, help symptoms and extend life while preserving quality of life. It is generally not given with an expectation of cure. Some people have prolonged benefit; others do not respond. Ask what success would look like for you: easier breathing, less pain, stable scans, more time or a balance of these.

Why are tumour tests important?

The team considers the cancer’s molecular results and PD-L1 result alongside your health. EGFR and ALK alterations can point towards a different treatment approach. PD-L1 is a protein involved in immune signalling; its result helps inform choices but is not a guarantee of benefit. This combination has evidence across PD-L1 groups, including low expression. Ask whether all relevant tumour testing is complete before your plan is finalised.

What are the alternatives?

The right comparison depends on tumour tests, symptoms, fitness and your preferences. Ask whether targeted therapy, immunotherapy alone, a different chemotherapy approach, a clinical trial or symptom-focused care is relevant. Supportive and palliative care can accompany active cancer treatment: help with breathlessness, pain, appetite and family concerns is not reserved for the last days of life.

Why this treatment is offered

Adding pembrolizumab to chemotherapy helped people in the study live longer and delayed cancer progression. Your benefit depends on your cancer and health.

CHAPTER 02

Your schedule, tests and appointments

IN THIS CHAPTER

Your treatment stages, checks and practical preparations, from before treatment through follow-up.

Treatment → review → next steps

Treatment stages

01 **Induction · cycles 1–4**
Three medicines

02 **Maintenance · after induction**
Two medicines

Your team confirms the active stage and dates. The full treatment guide explains each step.

This shows treatment timing, not guaranteed symptom-free days. Your team confirms the active phase and dates.

THREE MEDICINES
Induction · cycles 1–4

C1 C2 C3 C4

All three medicines on Day 1, once every 21 days, for up to four cycles. Your team may adjust the plan.

TWO MEDICINES
Maintenance · after induction

M1 M2 M3 …

Carboplatin stops. Pembrolizumab and pemetrexed may continue every 21 days while appropriate. Duration and changes are agreed with your oncology team.

Your team confirms each stage and discusses any changes with you.

BASELINE → TREATMENT → FOLLOW-UP

Treatment checks across the course

Relative stages, not equally spaced dates. Agree who books each check, reviews results and gives advice if a check is missed.

The team arranges and reviews tests. This guide shows expected timing; it does not record completed appointments. NCCP 00568 v5 · official regimen

THE WHOLE TREATMENT JOURNEY

Pembrolizumab + pemetrexed + carboplatin: treatment and checks

First-line metastatic non-squamous NSCLC without EGFR or ALK alterations

Induction: all three medicines on Day 1 every 21 days for up to four cycles. Maintenance: carboplatin stops; pembrolizumab and pemetrexed may continue every 21 days while appropriate.

Read from left to right. Columns show stages, not equal lengths of time. This is a general plan: no dates or personal information are needed. Your team confirms which steps apply and arranges appointments.

Treatment, tests and reviews across the course

Care / check	Before starting	First treatment	During treatment	Phase / dose review	After treatment
Treatment NCCP 00568	Confirm exact indication and planned course	Start the source-specific schedule	Continue the active phase as prescribed	Confirm hold, restart, maintenance or completion	Agree follow-up and next decision
Bloods and symptoms NCCP 00568	Blood count, kidney/liver, glucose, thyroid and viral checks; renal estimate	Folate and B12 in place before pemetrexed	Blood count, kidney/liver and glucose each cycle; thyroid every 3–6 weeks	After up to four cycles: carboplatin ends; continue maintenance checks	Folic acid for 21 days after last pemetrexed; remain alert to delayed immune symptoms
Treatment review NCCP 00568	Confirm exact cancer setting and biomarkers	Explain expected benefit and how it will be assessed	Report new symptoms between visits	Review response, tolerability and active medicines	Agree the next treatment or surveillance plan
Vitamins and steroids NCCP 00568	Folic acid lead-in and B12 in preceding week	One reconciled steroid prescription	Daily folate; B12 every three cycles while pemetrexed continues	Keep vitamins and rash prophylaxis in maintenance; review antiemetics	Folate continues for 21 days after final pemetrexed
Platinum effects NCCP 00568	Baseline nerve/hearing and renal assessment	Check kidney function and carboplatin prescription	Nerve/hearing review before each platinum cycle	No carboplatin in maintenance; follow any persisting effects	Follow unresolved kidney, hearing or nerve effects

What each check is for

1 Before starting

Before first dose

Confirm suitability and baseline safety

Ask / **act:** Confirm results, prescriptions and responsible teams.

2

Whole-course and completion review

At phase change or completion

Some checks and support continue after a drug stops

Ask / **act:** Provide the ongoing monitoring and urgent-contact plan.

At the next review, ask

- Which treatment phase am I in, and what happens next?
- Which test is expected next, who arranges it, and who explains the result?
- Who should I contact if an expected appointment has not arrived?

This map does not show whether a test has been ordered, booked or completed. Do not wait for a scheduled test if you develop new symptoms.

Practical treatment planning

Agree the arrangements with your oncology team.

STEP 1

Treatment phases and appointments

STEP 2

Checks after treatment

Folic acid continues throughout pemetrexed and for 21 days after its last dose. Vitamin B12 remains due every three cycles while pemetrexed continues. Immune-related symptoms can arise during or after treatment; maintain alert-card and urgent-contact education. Response imaging follows the consultant plan.

STEP 3

Access and practical support

intravenous (IV) treatment; no 46-hour home fluorouracil pump. Venous access choice is individual. Check B12 appointments, daily folate supply, steroid directions and the immunotherapy alert card across both phases.

Before the first infusion

Use this as a conversation checklist. Your own hospital's instructions and prescription take priority.

Tell the team your starting point

Describe your usual breathing, bowel pattern, energy, skin symptoms and ability to manage daily tasks. Mention autoimmune illness, organ transplant, previous lung inflammation, infections and all regular medicines. Changes are easier to recognise when the team knows what is normal for you. Do not minimise symptoms because you are worried treatment will be delayed.

Make the first week manageable

Have the oncology and out-of-hours numbers available, a working thermometer, your medicine schedule and a way to get urgent help. Ask who to ring if you miss a supportive medicine. Plan transport and practical help if needed. Bring your questions and, if you wish, someone you trust to listen and take notes.

During the infusion visit

Expect a symptom review and checks before drugs are released. Treatment is given through a cannula or another agreed venous access device. Tell the nurse immediately about burning or swelling around the drip, itching, flushing, dizziness or breathing changes. Waiting, pharmacy preparation and observation add to the time spent receiving the medicines.

What your tests and scans are checking

Different checks answer different questions: whether treatment is safe today, whether a symptom needs investigation, and whether the cancer is responding.

Blood counts: three separate jobs

Neutrophils help fight infection; a low level can make infection dangerous. Haemoglobin carries oxygen; a low level can contribute to tiredness and breathlessness. Platelets help stop bleeding; low levels can cause bruising or bleeding. Feeling well does not prove your counts are normal, and a previous normal result does not rule out a new problem.

Kidney, liver and hormone checks

Kidneys remove medicines and influence carboplatin dosing. Liver tests can show injury before obvious symptoms develop. Pembrolizumab can disturb hormone-producing glands, so blood glucose and thyroid monitoring matter too. Symptoms can prompt additional tests. Always report how you feel: routine blood tests alone do not exclude every immune side effect.

Scans and the treatment decision

Your team sets the scan schedule and compares scans with symptoms and examination findings. Stable disease can be a worthwhile result. New symptoms need review even before the next scan. An unusual scan pattern sometimes needs clarification; do not assume that apparent growth is simply an immune response or that feeling worse means treatment is working.

Why a delay or a change can be appropriate

A pause may allow recovery or investigation. Chemotherapy doses can sometimes be reduced. Pembrolizumab is generally held or stopped for relevant toxicity rather than dose-reduced. Ask which drug is changing and why; stopping one drug does not automatically mean stopping every part of the plan. Long-term treatment decisions belong in a discussion about benefit, risk and what matters to you.

CHAPTER 03

Symptoms and supportive care

IN THIS CHAPTER

Recognising changes, knowing when to call, and using your prescribed supportive-care plan.

Day 1 → Day 21: a general pattern

This is a visual orientation, not a prediction of your personal cycle. It shows when some effects often emerge and which effects are **not** tied to a particular day.

Treatment or check	Timing in this view
Treatment	 **Day 1** Day 1 · infusion visit
Nausea / appetite	 **Days 1–5** Early days · may continue longer
Blood counts / infection	 **Days 1–21** Counts may fall between visits · timing varies
Energy / mouth / bowels	 **Days 1–21** Symptoms vary across the cycle
Immune effects	 **Days 1–21** Any day · also after treatment ends

Induction: up to four cycles: Pembrolizumab, pemetrexed and carboplatin are given on Day 1 of a 21-day cycle for up to four cycles. The team may delay or change treatment after reviewing symptoms and test results.

Symptoms do not follow a fixed calendar: Chemotherapy effects may overlap between cycles. Immune-related effects can begin during treatment or after it has ended. A symptom band does not identify a safe day or predict your personal recovery.

Maintenance uses two medicines: After induction, carboplatin stops. Pembrolizumab and pemetrexed may continue every 21 days according to benefit, side effects and the consultant plan. Blood checks, folic acid, vitamin B12 and pemetrexed rash prophylaxis continue.

Care continues after the last infusion: Folic acid continues for 21 days after the final pemetrexed dose. Keep your immunotherapy alert information and follow the team's monitoring and urgent-contact plan.

Why these bands stay broad: The treatment bands show a reference 21-day cycle. Induction and maintenance contain different medicines. Symptom bands are broad educational illustrations, not measured daily risks or a prediction of when you will feel well; immune-related effects can occur after treatment ends. Band length does not show likelihood or severity, and the chart cannot tell you whether it is safe to receive treatment.

There are no guaranteed "safe days". Fever, shivers, sudden illness or new serious symptoms need prompt contact using your oncology team's instructions. New breathlessness, cough, severe diarrhoea, marked weakness or other possible immune-related symptoms can occur during or after treatment. Do not wait for the next appointment or use the timeline to decide a symptom is safe.

Between-treatment notes

Day 1: Your infusion day

Your team checks symptoms and blood results before treatment. Tell your nurse immediately about pain at the drip site, itching, flushing or breathing changes. The time in hospital includes checks and preparation, not just infusion time. Before leaving, check your take-home medicines, contact details and next appointment.

Day 2–5: The first days at home

Feeling sick can start on treatment day or later. Appetite, taste, sleep or bowel habits may change. Take prescribed supportive medicines as directed; ask for help if they are not controlling symptoms. Contact the team if vomiting persists, you cannot keep fluids down, or you develop new diarrhoea. Do not assume diarrhoea is ordinary chemotherapy upset.

Day 6–14: Keep checking how you feel

Blood counts may be reduced between treatments. You cannot tell your count from the date or from how well you feel. Fatigue or mouth symptoms may affect eating and daily activity. Use the urgent advice below for fever, shivering or suddenly feeling unwell, including with a normal temperature.

Day 15–21: Prepare for your next review

Some symptoms may settle; others can continue or appear for the first time. Improvement does not prove your blood counts have recovered. The next treatment depends on your review and tests. Bring your symptom record and questions. Confirm the next appointment: the next Day 1 is usually 21 days after this cycle's Day 1, if treatment proceeds.

Support around the medicines

1 

Vitamins and medicines at home

Take folic acid as prescribed and attend for vitamin B12 injections. Follow the written steroid and anti-sickness plan. Ask the team to explain any overlapping instructions.

2

When treatment changes

When carboplatin stops, the other medicines may continue. Vitamins and prescribed rash-prevention steroids are still needed with pemetrexed. Keep your immunotherapy alert card and report new symptoms promptly.

Side effects: what they mean and what to do

These are selected possibilities, not a prediction or a complete medicine leaflet. You do not need to work out the cause before asking for help.

Nausea, vomiting and appetite change

Medicines can stimulate sickness pathways in the gut and brain. Take preventive anti-sickness medicines as prescribed, even when you initially feel well. Smaller meals and frequent sips may be easier. Report poor control early so the plan can be adjusted. Persistent vomiting, inability to keep fluids down or much less urine need prompt advice. Do not wait until you are very dehydrated.

Fatigue: common, but not always “just treatment”

Fatigue may reflect treatment, disturbed sleep, reduced intake, anaemia or another illness. Balance rest with manageable movement and prioritise activities that matter to you. A sudden change, severe weakness, dizziness, breathlessness or inability to manage basic tasks needs assessment; hormone problems and other complications can resemble ordinary tiredness. Tell the team how fatigue affects daily life, not just that you feel tired.

Mouth soreness, ulcers and taste

Treatment can injure the lining of the mouth. Use a soft toothbrush, gentle mouth care and the rinses recommended by your team. Choose softer foods and avoid foods that sting. Ask for pain relief early. Report ulcers, white patches, painful swallowing or reduced intake; difficulty drinking or fever needs urgent advice. Arrange dental care with the oncology team before procedures.

Diarrhoea or abdominal pain

Pemetrexed can affect the bowel, but pembrolizumab can also cause bowel inflammation (colitis). Report new diarrhoea promptly, including how many stools above your normal pattern, blood or mucus, night-time symptoms and pain. Do not assume it is a harmless expected effect or rely on anti-diarrhoeal medicine without the team's instructions. Severe pain, bleeding or dehydration needs urgent assessment.

Constipation

Anti-sickness drugs, opioid painkillers, less food and less movement can slow the bowel. Ask about a bowel plan if you are prone to constipation or take opioids. Keep track of your usual pattern. Increasing abdominal pain, vomiting, a swollen abdomen or inability to pass wind needs prompt assessment rather than simply increasing laxatives.

Bruising, bleeding or breathlessness

Chemotherapy may reduce platelets or red cells. Report unexpected bruises, nosebleeds, bleeding gums or increasing breathlessness. Heavy bleeding, black stools with weakness, severe chest pain or severe breathing difficulty need emergency help. Do not start iron or alter blood thinners without review: the cause and treatment need to be established.

Skin, hair, nerves and hearing

Skin can become dry, itchy or develop a rash; report new changes, especially if spreading. Hair may thin. Tingling, numbness or hearing changes should be reported before the next dose because they may affect the plan. Blisters, peeling skin or sores involving the eyes or mouth need urgent assessment. Use gentle skin care and discuss suitable sun protection.

Pembrolizumab: symptoms that deserve special attention

Immune treatment can cause inflammation in healthy organs, during treatment or after it ends. Keep your immunotherapy alert card and tell other healthcare professionals you have received pembrolizumab.

Lungs, bowel, liver and kidneys

New cough or breathlessness, diarrhoea, yellow skin or eyes, dark urine, reduced urine or swelling can signal inflammation. These symptoms also have other causes, including infection and cancer. Contact the team promptly for assessment rather than trying to identify the organ or diagnosis yourself.

Hormones, heart and nervous system

Report unusual weakness, persistent headache, dizziness, marked thirst, frequent urination, palpitations, muscle weakness or changes in vision. Severe chest pain, collapse, confusion, seizures or major breathing difficulty require emergency care. Some hormone effects need ongoing replacement treatment even after immunotherapy stops.

Why early reporting matters

The team may pause treatment and investigate. Some reactions need steroids or other immune-suppressing treatment; others need hormone replacement. These are different from the short planned steroid course around chemotherapy. Do not start leftover steroids, change their dose or stop a prescribed course yourself. Carry the treatment name when attending emergency care.

Living with treatment

The aim is a plan that fits your life as well as your treatment. Bring practical concerns to the team early.



Food, fluids and weight

Aim for enough food and fluid to maintain strength, using foods you tolerate. Tell the team about weight loss, swallowing problems or difficulty shopping or cooking; ask for dietitian support. A fixed fluid target is not right for everyone, particularly with heart or kidney disease. “Anti-cancer” diets, fasting and high-dose supplements are not substitutes for treatment and can create nutritional or interaction problems.

Movement, work and rest

Try activity that feels manageable and increase or reduce it according to symptoms. Short walks or simple daily tasks may be more realistic than your usual exercise. Discuss work adjustments and travel plans. Do not drive if dizzy, unusually tired or affected by sedating medicines. Sudden deterioration should prompt assessment, not an attempt to push through exercise.

Infection precautions without isolation

Wash hands, practise food hygiene and avoid close contact with people who are unwell. Ask about vaccinations and their timing; some vaccines are unsuitable during treatment. You can usually still spend time with family. Keep your urgent contact plan available: precautions reduce risk but do not make fever or sudden illness safe to watch at home.

Intimacy, fertility and pregnancy

Discuss fertility preservation before treatment if relevant. Treatment may harm a developing baby; ask for a written contraception plan covering all three medicines and the period after the last dose. Do not breastfeed during treatment; confirm when it would be safe to restart. Tell the team promptly about a possible pregnancy. Ask about barrier protection and handling body fluids under your hospital’s advice.

Emotional support and choices

It is reasonable to feel worried, angry, uncertain or overloaded. Ask for psycho-oncology, counselling, social work or family support. You can request information in smaller pieces, an interpreter or a written summary. Tell the team what you most want to preserve in daily life. You can revisit treatment decisions and ask for more time or another explanation.

A short treatment dictionary

Terms you may hear in clinic.

Cycle · induction · maintenance

A cycle is a planned treatment-and-interval period. Induction is the initial combination phase. Maintenance is ongoing treatment after that phase; it does not mean side-effect-free treatment.

Response · stable disease · progression

Response means the cancer has reduced by the assessment used. Stable disease means it has not changed enough to meet response or progression criteria. Progression means growth or spread meeting the assessment criteria. Your team interprets these alongside how you are feeling.

Neutropenia · anaemia · thrombocytopenia

Neutropenia means low infection-fighting neutrophils. Anaemia means low haemoglobin. Thrombocytopenia means low platelets. These are different results with different implications; one normal number does not establish that all treatment risks are low.

Immune-related · colitis · pneumonitis

Immune-related effects come from immune activity against healthy tissue. Colitis means inflammation of the colon; pneumonitis means inflammation in the lungs. A symptom alone cannot establish either diagnosis.

Questions worth asking

How will we know if it is helping?

Ask your oncologist what the treatment is aiming to achieve for you, when scans are planned and what would lead to continuing, changing or stopping it. How you feel and the amount of side effects do not reliably show how well treatment is working.

Why might a cycle be delayed?

A delay can give time to recover or investigate a new symptom. It is a safety decision, not a personal failure. Ask which medicines are affected, what needs to improve and when you will be reviewed.

Will I get every side effect?

No. The body map shows selected possibilities, not a prediction. Ask about common effects, serious less common effects, and any long-term risks that matter particularly to you. The same symptom can have several causes.

What should I organise before starting?

Confirm blood-test and Day Ward arrangements, transport home, your medicines schedule and a thermometer. Keep the oncology contact instructions and immunotherapy alert card available. Ask for an interpreter or accessible information if useful.

What else belongs in the consent conversation?

Your priorities, expected benefit, alternatives including supportive care, fertility, contraception, pregnancy or breastfeeding, work, driving and the possibility of lasting side effects. You can ask more questions or reconsider treatment at any stage.

Do more side effects mean the treatment is working better?

No. Side effects are not a reliable personal measure of cancer response. Report them so they can be treated; do not tolerate serious symptoms in the hope that they are a sign of benefit.

Will the same three medicines continue indefinitely?

No. The planned carboplatin phase is up to four cycles. Pembrolizumab and pemetrexed may continue afterwards. Ask about the intended duration of each drug, review points and reasons for a change.

Can I use this page to decide whether my blood tests are good enough?

No. Your treating team combines blood results, symptoms, prior toxicity and the prescribing protocol. This page explains the checks; it does not authorise a dose or a treatment decision.

What should I say when calling about a symptom?

Say you are receiving pembrolizumab, pemetrexed and carboplatin (or your current maintenance drugs), the last treatment date, what has changed, when it started, your temperature and whether you can eat, drink and pass urine. Say if symptoms are worsening. Do not delay calling to collect every detail.

> **Try explaining the plan back:** Which medicines are you having now? What changes after induction? What would make you call today? Ask your team to fill any gaps.

Supportive prescription draft for review. This is a regimen template. It is not an issued patient prescription; individual review and completion are required.

Induction: carboplatin-containing cycles 1–4

Active medicines: pembrolizumab + pemetrexed + carboplatin

Emetogenic risk: high

Medicine	Dose / route	When
Aprepitant	125 mg orally Day 1; 80 mg orally Days 2–3	Before systemic anticancer therapy (SACT) on Day 1, then daily on Days 2–3
Ondansetron	16 mg orally	Day 1 Adjust for QT/individual risk.
Antiemetic dexamethasone reference	12 mg orally Day 1; 8 mg orally daily Days 2–4	Dedicated carboplatin AUC ≥4 NCIS V8 row Reference course only: reconcile with pemetrexed rash prophylaxis and immunotherapy context before a single prescription is issued.
Metoclopramide	10 mg orally three times daily as prescribed	Subsequent days NCIS carboplatin row does not specify an end day or PRN qualifier. Prescriber/ pharmacy must specify duration and suitability.
Folic acid	350–1000 micrograms orally daily	At least five doses in seven days before first pemetrexed; throughout treatment and 21 days after the final dose
Vitamin B12 (hydroxocobalamin)	1000 micrograms intramuscular (IM)	Week before first pemetrexed and every three cycles thereafter Later doses may be on the pemetrexed day.

Medicine	Dose / route	When
Pemetrexed rash prophylaxis: dexamethasone equivalent	4 mg orally twice daily	Day before, day of and day after pemetrexed This is the regimen rash schedule; prescriber/pharmacy must reconcile it with antiemetic dexamethasone into one unambiguous prescription, including aprepitant interaction. Do not simply add both courses.

Additional phase-specific support

- High emetic risk is driven by carboplatin AUC5. Dedicated NCIS carboplatin row has no routine olanzapine.
- Do not combine rash and antiemetic dexamethasone courses without reconciliation.
- Carry immunotherapy alert card; do not wait until next cycle for possible immune toxicity.

Before signing: verify allergies, current medicines, interactions, renal / hepatic function, QT risk, sedation, prior response, total steroid exposure and the rescue-medicine duration. Do not duplicate medicines from another phase.

Individual changes / rescue duration: _____
Prescriber / date: _____ Pharmacy review: _____

Maintenance: pembrolizumab + pemetrexed

Active medicines: pembrolizumab + pemetrexed

Emetogenic risk: low

Medicine	Dose / route	When
Folic acid	350–1000 micrograms orally daily	At least five doses in seven days before first pemetrexed; throughout treatment and 21 days after the final dose
Vitamin B12 (hydroxocobalamin)	1000 micrograms intramuscular (IM)	Week before first pemetrexed and every three cycles thereafter Later doses may be on the pemetrexed day.

Medicine	Dose / route	When
Pemetrexed rash prophylaxis: dexamethasone equivalent	4 mg orally twice daily	Day before, day of and day after pemetrexed This is the regimen rash schedule; prescriber/pharmacy must reconcile it with antiemetic dexamethasone into one unambiguous prescription, including aprepitant interaction. Do not simply add both courses.
Low-risk antiemetic / rescue	Individual prescription	Before treatment or as required according to current NCCP/local plan Account for the existing rash-prophylaxis steroid. No automatic continuation of induction aprepitant or high-risk multi-day steroid course.

Additional phase-specific support

- Folate, B12 and rash prophylaxis continue because pemetrexed continues.
- Carboplatin ends after induction; review supportive prescriptions at the phase change.
- Continue renal, glucose, thyroid and immune-toxicity monitoring.

Before signing: verify allergies, current medicines, interactions, renal / hepatic function, QT risk, sedation, prior response, total steroid exposure and the rescue-medicine duration. Do not duplicate medicines from another phase.

Individual changes / rescue duration: _____
Prescriber / date: _____ Pharmacy review: _____

What to watch for

You do not need to know which medicine caused a symptom before calling.

1 Hormones / brain

Pembrolizumab

Unusual tiredness, dizziness, headache, thirst or passing more urine.

2 Lungs

Pembrolizumab

New cough or breathlessness: report promptly. Infection, cancer or a clot can cause similar symptoms.

3 Heart

Pembrolizumab

Chest pain, racing heartbeat or fainting needs urgent assessment.

4 Liver

Pembrolizumab

Yellow eyes or skin, dark urine or changes in blood tests.

5 Kidneys

Pemetrexed / pembrolizumab

Kidney function is monitored. Report less urine or new swelling.

6 Bowel

Pembrolizumab / chemotherapy

New diarrhoea, persistent vomiting or severe tummy pain: contact your team.

7 Skin / nerves

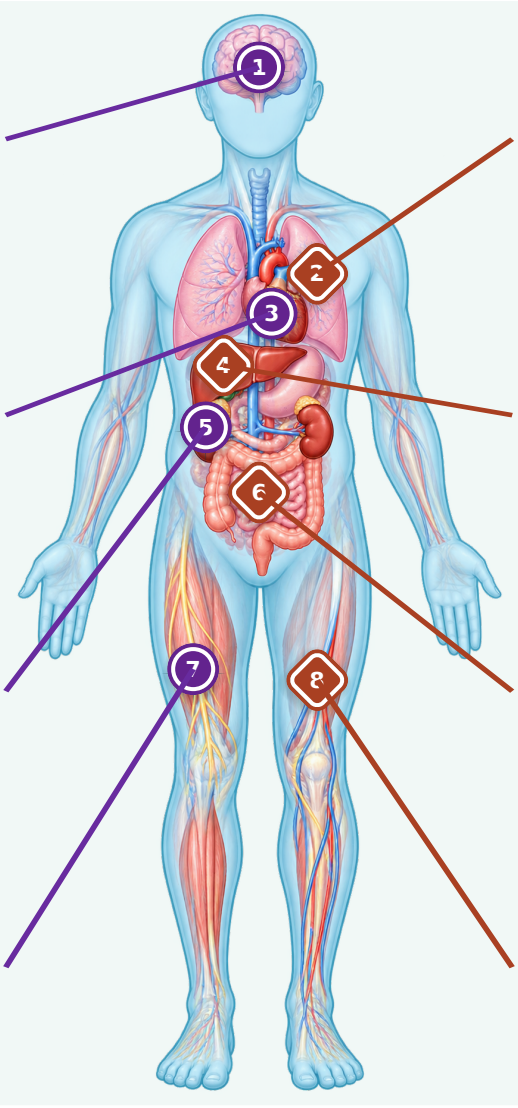
Pembrolizumab / chemotherapy

Spreading rash, blisters, weakness or new numbness should be reported.

8 Blood / infection

Pemetrexed / carboplatin

Fever, shivers or suddenly feeling unwell: contact oncology immediately.



Selected effects, not every possible effect or a prediction. Blood-count, skin and line markers are schematic. Number and text carry meaning; colour is decorative.

Contact the oncology team promptly for fever or feeling acutely unwell, new breathlessness or cough, severe diarrhoea, dehydration, chest pain, confusion, severe rash or other new symptoms. Use the emergency contact instructions given by your team.

When to get help

Call 112 / 999 for severe breathing difficulty, severe chest pain, collapse or a seizure.

Call oncology immediately: temperature 37.5°C or above, 36°C or below, shivering or suddenly feeling unwell, even with a normal temperature. Follow your hospital's temperature instructions. If you cannot reach the team, seek urgent medical assessment. Do not wait for the next appointment.

Oncology daytime: 087 12345678 · **Out of hours:** 087 87654321

EXAMPLE ONLY — team must replace both numbers before patient use.

CHAPTER 04

Sources and further information

IN THIS CHAPTER

The references behind this information and addresses you can use to find further help.

Use this information alongside the plan agreed with your treatment team. Your team confirms your medicines, doses and appointments.

1. **National Cancer Control Programme (NCCP) 00568 v5 · official regimen**
   https://healthservice.hse.ie/documents/6891/568_V5_Pembrolizumab_PEMEtrexed_CARBOplatin.pdf
2. **Macmillan · pembrolizumab and immune effects**
   https://www.macmillan.org.uk/cancer-information-and-support/treatments-and-drugs/pembrolizumab
3. **Cancer Research UK · pemetrexed mechanism**
   https://www.cancerresearchuk.org/about-cancer/treatment/drugs/pemetrexed
4. **National Cancer Institute · carboplatin mechanism**
   https://www.cancer.gov/about-cancer/treatment/drugs/carboplatin
5. **Macmillan · pemetrexed and carboplatin (two-drug information)**
   https://www.macmillan.org.uk/cancer-information-and-support/treatments-and-drugs/pemetrexed-and-carboplatin
6. **National Cancer Institute · early and delayed nausea**
   https://www.cancer.gov/about-cancer/treatment/side-effects/nausea-vomiting
7. **HSE · chemotherapy safety and when to contact your team**
   https://www2.hse.ie/conditions/chemotherapy/side-effects/
8. **KEYNOTE-189 · five-year randomised trial follow-up (2023)**
   https://pubmed.ncbi.nlm.nih.gov/36809080/
9. **KEYTRUDA · prescribing information**
   https://www.medicines.org.uk/emc/product/2498/smpc
10. **Alimta (pemetrexed) · prescribing information**
    https://www.medicines.org.uk/emc/product/7443/smpc
11. **eviQ · patient information for this three-drug combination**
    https://www.eviq.org.au/medical-oncology/respiratory/non-small-cell-lung-cancer-advanced-metastatic/3510-nsclc-metastatic-carboplatin-pemetrexed-and-p/patient-information
12. **NCI · immunotherapy and organ inflammation**
    https://www.cancer.gov/about-cancer/treatment/side-effects/organ-inflammation
13. **NCI · fatigue and cancer**
    https://www.cancer.gov/about-cancer/treatment/side-effects/fatigue
14. **NCI · mouth and throat problems**
    https://www.cancer.gov/about-cancer/treatment/side-effects/mouth-throat
15. **NCI · infection and neutropenia**
    https://www.cancer.gov/about-cancer/treatment/side-effects/infection
16. **NCI · biomarker testing for cancer treatment**
    https://www.cancer.gov/about-cancer/treatment/types/biomarker-testing-cancer-treatment
17. **NCI · palliative care alongside cancer treatment**
    https://www.cancer.gov/about-cancer/advanced-cancer/care-choices/palliative-care-fact-sheet
18. **KEYNOTE-189 five-year update**
    https://pmc.ncbi.nlm.nih.gov/articles/PMC10082311/
19. **NCCP emetogenic classification V6**
    https://healthservice.hse.ie/documents/7152/Classification_Document_for_Systemic_Anti-_Cancer_Therapy_systemic anticancer therapy (SACT)_Induced_Nausea__pWgDCrm.pdf
20. **NCCP NCIS Medical Oncology antiemetics V8, Table 1**
    https://healthservice.hse.ie/documents/7180/Supportive_Care_Antiemetics_for_inclusion_NCIS_Medical_Oncology.pdf